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Exercise of Structure and Stellar Evolution

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Problem sheet 1

Exercise 1

Calculate the degree of ionisation of H at the centre of the sun.

1. if all the density is due to H.
2. if the composition is $X = 0.35$, $Y = 0.65$.

What does the result suggest you?

Solution

If I want to calculate the degree of ionization in the center of the sun I need to use the Saha Equation:

$$\frac{n_{i+1}}{n_i} = \frac{g_{i+1}}{g_i} \frac{2}{n_e} \frac{(2\pi m_e)^{3/2}}{h^3} (kT)^{3/2} \exp\left(-\frac{\chi_i}{kT}\right) \quad (1)$$

where n_e is the number density (per unit volume) of free electrons (that is, electrons that are not attached to any atom) and m_e is the electron mass. The quantities g_i and g_{i+1} are the statistical weights of the two ionization stages, and χ_i is the energy required to ionize an atom in the ground state of stage i to the ground state of stage $i + 1$.

First of all we need the temperature and the density of electrons at the center of the sun, which we can take respectively equal to $T \approx 15.7 \cdot 10^6$ K and $n_e \approx 6.1 \cdot 10^{31} \text{ m}^{-3}$. Next, I need to figure out what to assign to the statistical weights. For hydrogen it is easy because it turns out that for hydrogen, the statistical weights are given by $g_n = 2n^2$. Assuming most of the neutral hydrogen is in the ground state, we can make the approximation $g_1 = 2$ (here, g_1 is the statistical weight of the ground state). For the ionized hydrogen, $g_{i+1} = 1$ because the ionized state is just a proton. Finally, the ionization energy of hydrogen is $\chi = 13.6$ eV. So, plug all together I obtain:

$$\frac{n_H^1}{n_H^0} = \frac{1}{n_e} \frac{(2\pi m_e)^{3/2}}{h^3} (kT)^{3/2} \exp\left(-\frac{13.6}{kT}\right) \approx 2.4 \quad (2)$$

So I get for the fraction of ionized hydrogen atoms with the above values:

$$x_H^1 = \frac{n_H^1}{n_H} = \frac{n_H^1/n_H^0}{1 + n_H^1/n_H^0} \approx 0.7 \quad (3)$$

with $n_H = n_H^0 + n_H^1$. Now let's move on to the case where there is 35% hydrogen and 65% helium.

Now I have six types of particles: neutral and ionized hydrogen; neutral, ionized, and double ionized helium; and electrons. There are three types of ionization energy: χ_H^0 H for

hydrogen and χ_{He}^0, χ_{He}^1 for neutral and single ionized helium ($\chi_H^0 = 13.6$ eV, $\chi_{He}^0 = 24.58$ eV and $\chi_{He}^1 = 54.4$ eV). Each ionized hydrogen atom contributes the energy χ_H^0 to the internal energy, each helium atom in the first stage of ionization the energy χ_H^0 and each helium atom completely stripped of its two electrons the energy $\chi_{He}^0 + \chi_{He}^1$. By $\chi_H^0, \chi_H^1, \chi_{He}^0, \chi_{He}^1, \chi_{He}^2$ we define degrees of ionization, i.e. x_i^r gives the number of atoms of type i in ionization state r ($= r$ electrons lost) divided by the total number of atoms of type i (irrespective of their state of ionization):

$$x_H^1 = \frac{n_H^1}{n_H} \quad x_{He}^1 = \frac{n_{He}^1}{n_{He}} \quad x_{He}^2 = \frac{n_{He}^2}{n_{He}}$$

with $n_H = n_H^0 + n_H^1$ and $n_{He} = n_{He}^0 + n_{He}^1 + n_{He}^2$ where the n_i^r are number densities of ions of type i in ionization state r . From the Saha equation it's possible to calculate the fraction as I done in the previous part.

- **Hydrogen:** $n_H^1/n_H^0 = 2.41$ (from the previous calculation)
- **Helium-1:** the first ionization of helium occurs by $\chi_{He}^1 = 24.48$ eV. For the first helium ionization the ratio of statistical weights is equal to $g_1/g_0 = 2$. For helium-1 it is found that $n_{He}^1/n_{He}^0 \approx 9.67$.
- **Helium-2:** the second ionization of helium occurs by $\chi_{He}^2 = 54.4$ eV. For the second helium ionization the ratio of statistical weights is equal to $g_2/g_1 = 1/2$. For helium it is found that $n_{He}^2/n_{He}^1 \approx 2.36$.

So, the degree of ionization are:

$$\begin{aligned} x_H^1 &= \frac{n_H^1/n_H^0}{1 + n_H^1/n_H^0} \approx 0.70 \\ x_{He}^1 &= \frac{n_{He}^1/n_{He}^0}{1 + n_{He}^1/n_{He}^0 + n_{He}^2/n_{He}^1} \approx 0.74 \\ x_{He}^2 &= \frac{n_{He}^2/n_{He}^1}{1 + n_{He}^1/n_{He}^0 + n_{He}^2/n_{He}^1} \approx 0.18 \end{aligned} \tag{4}$$

Now we simply need to add all these contributions weighted on the individual percentages. That is:

$$E = \left[Xx_H^1 + \frac{1}{4}Y(x_{He}^1 + 2x_{He}^2) \right] \mu_0 = 0.83 \tag{5}$$

where

$$\mu_0 = \left(\frac{X}{\mu_H} + \frac{Y}{\mu_{He}} \right)^{-1} \approx \frac{80}{41} \tag{6}$$

The fact that the degree of ionization is higher in the case of hydrogen and helium compared to hydrogen alone suggests that the introduction of helium has influenced the ionization process in such a way as to increase the overall degree of ionization of the gas. In reality, this result was expected since inside the sun, since the centre of sun is completely ionized. Therefore, if you take more truthful fractions of particles (and certainly taking only hydrogen is not truthful) you have a greater degree of ionization.

This could be due to several factors, including:

1. **Effect of Helium on Temperature and Density:** The addition of helium may influence the temperature and density of the gas, thereby altering the ratio between ionization levels and available energy states. This could lead to a higher likelihood of ionization.
2. **Particle Interactions:** Interaction between hydrogen and helium particles could affect the ionization process, for example, by providing additional energy to overcome energy barriers.
3. **Atomic Resonances:** Helium may have energy levels that overlap with those of hydrogen, creating atomic resonances that favor the ionization process.
4. **Shielding Effects:** Helium, being an inert gas, may partially shield hydrogen electrons from the effects of the nucleus, allowing them to ionize more easily.

In general, the effect of helium on ionization will depend on the temperature, density, and composition of the gas mixture, as well as the specific physics of the gases involved.

Exercise 2

In a star its nucleus has reached a density of $3 \cdot 10^5 \text{ gr/cm}^3$. Answer the following questions:

- What is the electron density, if the core is constituted by H?
- What is the p_F of the e^- gas?
- What is the p_F of the proton gas?
- What is the Fermi energy of both?
- At which temperature can we consider these electrons to be degenerate?
- At this temperature will the protons be degenerate too?

Solution

As the density increases, the number of electrons with a given momentum can exceed the exclusion condition. The electrons are then forced into higher momentum states, until they reach a threshold momentum: the Fermi momentum:

$$n_e dV = \int_0^\infty \frac{8\pi p^2}{h^3} dp dV = \frac{8\pi}{3h^3} p_F^3 dV \quad (7)$$

So, the electron density, in general, is defined as:

$$n_e = \frac{8\pi}{3h^3} p_F^3 \quad (8)$$

This relationship is very useful because it relates the electron density to the Fermi momentum. However, it is also possible to write that the numerical density of electrons is

nothing more than the density of the object taken into consideration (in this case the nucleus of a star) divided by its mass (in this case just hydrogen). So, I can write:

$$n_e = \frac{\rho}{\mu_e m_H} \quad (9)$$

Since I have only hydrogen, the average molecular weight is define has:

$$\mu_e = \left(\sum_i \frac{X_i Z_i}{\mu_i} \right)^{-1} = 1 \quad (10)$$

and $m_H = 1,00784$ u that is $m_H = 1.673 \cdot 10^{-24}$ g. So, the electron density is:

$$n_e = \frac{3 \cdot 10^5}{1.673 \cdot 10^{-24}} = 1.79 \cdot 10^{29} \text{ cm}^{-3} \quad (11)$$

Using Eq.(8) it is possible to calculate the Fermi moment, as:

$$p_F = \left(\frac{n_e 3h^3}{8\pi} \right)^{\frac{1}{3}} = 1.84 \cdot 10^{-22} \text{ kg m s}^{-1} \quad (12)$$

Since we are dealing with a nucleus of only hydrogen, then clearly the number of electrons will be equal to the number of protons, i.e. $n_e = n_p$. Consequently also their numerical density and therefore also the Fermi moment. To further clarify this aspect, it is sufficient to note that in all the formulas used the mass of the proton or electron never appears individually, therefore there is no dependence on the individual masses. Rather we have the mass of hydrogen. However this does not apply to the Fermi energy since the Fermi energy is defined as:

$$E_F = \frac{p_F^2}{2m_i} \quad (13)$$

where m_i in this case can only be either the mass of the electron or that of the proton. Knowing that $m_e \ll m_p$ then the Fermi energy $E_{F,e} \gg E_{F,p}$. Putting the numbers in, I get:

$$\begin{aligned} E_{F,e} &= 1.857 \cdot 10^{-14} \text{ J} = 1,159 \cdot 10^7 \text{ eV} \\ E_{F,p} &= 1.01 \cdot 10^{-17} \text{ J} = 6303 \text{ eV} \end{aligned} \quad (14)$$

To determine the temperature at which electrons can be considered degenerate, we compare the thermal energy (kT) with the Fermi energy of the electron. Here, k represents the Boltzmann constant, and T is the absolute temperature in Kelvin. If the thermal energy (kT) is much smaller than the Fermi energy, then the electrons can be considered degenerate. In other words, when the temperature is low enough such that the thermal energy is much smaller than the Fermi energy, quantum behaviors become dominant, and electrons occupy discrete quantum states rather than freely distributing over all available energy states. The approximate limit at which we can consider electrons as degenerate is when $kT \ll E_F$, where E_F is the Fermi energy. This limit defines the so-called Fermi temperature. The Fermi temperature is approximately equal to the Fermi energy divided by the Boltzmann constant. So, to determine the temperature at which electrons can be considered degenerate, we can use the relation:

$$T_F = \frac{E_F}{k} \quad (15)$$

Where T_F is the Fermi temperature, E_F is the Fermi energy, and k is the Boltzmann constant. In number:

$$T_{F,e} = 1.345 \cdot 10^9 K \quad (16)$$

At this temperature the protons are not degenerate. If we want to decide whether a gas of fermions is degenerate, then you would compare the temperature of the gas with its Fermi temperature. If $T \ll T_F$ then the gas can be considered completely degenerate. If $T \approx T_F$ then the gas is partially degenerate. If $T \gg T_F$ then the gas is not degenerate. In my case, since the Fermi temperature of the protons is $T_{F,p} = 7.3 \cdot 10^5$ K then $T_{F,e} \gg T_{F,p}$ the protons at this temperature are not degenerate

Problem sheet 2

Exercise 1

Suppose the Sun is a mono-atomic gas in hydrostatic equilibrium. Calculate its internal energy and compare it with the gravitational energy. Applying the *virial theorem*, find a lower limit to the mean temperature of the Sun (T weighted by mass) and compare the result with the temperature of the numerical model of the Sun.

Solution

The Euler's equation is defined as:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho} \nabla p - \nabla \Phi \quad (17)$$

where $\mathbf{u}$ is the velocity field and Φ the gravitational potential. In the hypothesis of **hydrostatic equilibrium** all the derivatives with respect to time vanish since the fluid is at equilibrium, furthermore, $\mathbf{u} = 0$ because the fluid is static. Then, from Euler's equation we obtain that:

$$\frac{1}{\rho} \nabla p = -\nabla \Phi \quad (18)$$

This equation is called **hydrostatic equilibrium equation** and implies that for a fluid to be in a condition of inertia, external and internal forces, including gravitational ones, must be balanced by pressure forces. I treat the sun as a sphere, therefore in spherical coordinates Eq. (18) became:

$$\frac{dP}{dr} = -\rho \frac{Gm}{r^2} \quad (19)$$

where the potential has clearly been made explicit. Sometimes it is more convenient to phrase this in terms of change per unit mass, so:

$$\frac{dP}{dm} = \frac{dP}{dr} \frac{dr}{dm} = \frac{dP}{dr} \left(\frac{dm}{dr} \right)^{-1} = \frac{dP}{dr} \frac{1}{4\pi r^2 \rho} = -\frac{Gm}{4\pi r^4} \quad (20)$$

Now we need a relation between pressure, internal energy and thus the gravitational energy. For do that, I will next derive a volume-integrated form of the equation of hydrostatic equilibrium: the virial theorem. The proof can be find in Appendix A. I can use the virial theorem to estimate the mean temperature inside the Sun. Let T be the Sun's mass-averaged temperature. The internal energy is therefore:

$$U = \frac{3}{2} M \frac{\mathcal{R}T}{\mu} \quad (21)$$

While the gravitational energy depends somewhat on the internal density distribution of the Sun, that in principle I don't know but it must be something like:

$$\Omega = -\alpha \frac{GM^2}{R} \quad (22)$$

where α is a constant of order unity that describes our ignorance of the internal density structure. Applying the virial theorem and solving, we obtain:

$$\frac{3}{2}M \frac{\mathcal{R}T}{\mu} = \frac{1}{2}\alpha \frac{GM^2}{R} \implies T = \frac{\alpha}{3} \frac{\mu}{\mathcal{R}} \frac{GM}{R} \quad (23)$$

If I plug in:

- $M = M_{\odot} = 2 \cdot 10^{30}$ kg,
- $R = R_{\odot} = 7 \cdot 10^8$ m,
- $\mu = \frac{1}{2}$ appropriate for a gas of pure, ionized hydrogen. This is related to the gamma of a $5/3$ monatomic gas,
- $\alpha = \frac{3}{5}$ appropriate for a uniform sphere,

I obtain that $U = -\Omega/2 \approx 2.28 \cdot 10^{41}$ J and $T = 2.3 \cdot 10^6$ K. It is obviously much hotter than the surface temperature of about 6000 K. Comparing to the result with the temperature of the numerical model of the Sun, that is $T = 10^6 - 10^7$ K they are clearly on the same order of magnitude.

Exercise 2

Consider a polytrope of index $n = 1$ as a model of a star of density 50 g/cm^3 , radius $8 \cdot 10^5$ km and chemical composition $X = 0.65$ and $Y = 0.35$.

- a) Write the structure equation of this star and solve it. Obtain the values of P , T and ρ at the centre and at $3 \cdot 10^5$ km from the centre.
- b) Calculate its mass and the gradients of P , T and ρ at the surface.

Solution

The structure equation are:

- The relation between mass and radius: Clearly, $m = \rho(4/3)\pi r^3$ then, $dm = 4\pi r^2 \rho dr$, and so:

$$\frac{dr}{dm} = \frac{1}{4\pi r^2 \rho} \quad (24)$$

- This is the hydrostatic equilibrium equation, which I just obtained a moment ago. It expresses the balance between gravitational force and pressure force:

$$\frac{dP}{dm} = -\frac{Gm}{4\pi r^4} \quad (25)$$

- Here we have the equation of energy production which is the third equation of stellar structure. The proof of this formula can be find in Appendix B.

$$\frac{dL}{dm} = q_{nuc} \quad (26)$$

- The fourth and final equation is basically a rewrite of the equation of radiative transport. The proof of the radiative transport equation is done in Exercise 4.

$$\frac{dT}{dm} = -\frac{3}{4ac} \frac{\kappa_R}{T^3} \frac{L}{(4\pi r^2)^2} \quad (27)$$

Here, κ_R is the Rosseland mean opacity.

Considering a polytropic sphere, the equation of state is:

$$P = K_P \rho^{\gamma_P} \quad (28)$$

where K_P is the adiabatic constant and adiabatic index, and γ_P is defined as: $\gamma_P = 1 + 1/n$. I can proceed to substitute it into the equation of hydrostatic balance, and after a little bit of simplification I find:

$$\frac{(n+1)K_P}{4\pi G n} \frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho^{(n-1)/n}} \frac{d\rho}{dr} \right) = -\rho \quad (29)$$

As a second-order equation, this ODE requires two boundary conditions. At the surface $r = R$, the pressure $P(R) = 0$, and since $P \propto \rho^{\gamma_P}$, one of our boundary conditions could be $\rho(R) = 0$. However, usually one instead sets the density at the center, $\rho(0) = \rho_c$, and then R is the radius at which ρ first goes to zero. For the second boundary condition, recall that the original equation of hydrostatic equilibrium read $dP/dr = -\rho G m/r^2$. Unless the density becomes infinite in the center, m must vary as $m \propto \rho_c r^3$ near the center of the star, so m/r^2 must vary as r . Thus dP/dr goes to zero at $r = 0$, and, for our polytropic equation of state, $d\rho/dr$ must therefore approach zero in the center as well. This is our second boundary condition: $d\rho/dr = 0$ at $r = 0$. If I make a change of variables. Let ρ_c be the density in the center of the star, and let us define the new variable θ by:

$$\rho = \rho_c \theta^n \quad (30)$$

where θ is a dimensionless number, and that it runs from 1 at the center of the star to 0 at the edge of the star. With this change of variable, we can re-write the equation of hydrostatic equilibrium as:

$$\left[\frac{(n+1)K_P}{4\pi G \rho_c^{(n-1)/n}} \right] \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\theta}{dr} \right) = -\theta^n \quad (31)$$

The quantity in square brackets has units of length squared, and this suggests a second change of variables. I let:

$$\alpha^2 = \left[\frac{(n+1)K_P}{4\pi G \rho_c^{(n-1)/n}} \right] \quad (32)$$

and I then set:

$$\xi = \frac{r}{\alpha} \quad (33)$$

so that ξ is also a dimensionless number. With this substitution, the equation becomes the Lane-Emden equation:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n \quad (34)$$

For $n = 1$ it's possible to find a exact analytical solution, that is:

$$\theta(\xi) = \frac{\sin(\xi)}{\xi} \quad (35)$$

Now, from here it's possible compute the mass of the star, as:

$$M = \int_0^R 4\pi r^2 \rho dr = 4\pi \alpha^3 \rho_c \int_0^{\xi_1} \xi^2 \theta^n d\xi = -4\pi \alpha^3 \rho_c \int_0^{\xi_1} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) d\xi \quad (36)$$

In conclusion I obtain:

$$M = -4\pi \alpha^3 \rho_c \xi_1^2 \left(\frac{d\theta}{d\xi} \right)_{\xi_1} \quad (37)$$

Plug the solution (35):

$$M = -4\pi \alpha^3 \rho_c \xi_1^2 \left(\frac{\cos(\xi_1)}{\xi_1} - \frac{\sin(\xi_1)}{\xi_1^2} \right) = 4\pi \alpha^3 \rho_c (\sin(\xi_1) - \xi_1 \cos(\xi_1)) \quad (38)$$

Now that I have the mass, I can find the density at the centre of star ρ_c . For do that, I know that the mean density is defined as $\bar{\rho} = M/V$ where $V = \frac{4}{3}\pi R^3$ so:

$$\bar{\rho} = \frac{3M}{4\pi \alpha^3 \xi_1^3} = \frac{3\rho_c}{\xi_1^3} (\sin(\xi_1) - \xi_1 \cos(\xi_1)) \quad (39)$$

The numerical solution with $n = 1$ and $\xi_1 = \pi$, the density at the centre is:

$$\rho_c \approx 165 \frac{\text{g}}{\text{cm}^3} \quad (40)$$

that is a reasonable value, Indeed, this central density resembles that at the center of the Sun, which is $\rho_{c,\odot} \approx 150 \frac{\text{g}}{\text{cm}^3}$. Knowing ρ_c it's possible to calculate the pressure at the centre. In particular: $P_c = K_P \rho_c^{(n+1)/n}$, so using the mass-radius relation to solve K_P , I obtain:

$$P_c = \frac{(4\pi G)^{1/n}}{n+1} \left(\frac{GM}{\sin(\xi_1) - \xi_1 \cos(\xi_1)} \right)^{\frac{n-1}{n}} \left(\frac{R}{\xi_1} \right)^{\frac{3-n}{n}} \rho_c^{\frac{n+1}{n}} \quad (41)$$

In the case $n = 1$, it becomes:

$$P_c \approx 7.31 \cdot 10^{17} \text{ Pa} \quad (42)$$

The most questionable part of what we have done so far is to assume that all polytropic stars with fixed n share the polytropic constant k , but why should this be so? In other words we ask ourselves how k varies as the mass of a star varies. Stellar mechanisms, for main sequence stars, are very similar. I know that hydrogen is burned into helium, then

helium into lithium and so on, forming layers of different atoms. In reality, it is a good approximation that the temperature inside the main sequence stars is more or less the same. From the state equation, $p = \frac{k_B}{\mu m_p} \rho T$, it's possible to find the temperature at the centre:

$$T_c = \frac{P_c \mu m_p}{\rho_c k_B} \approx 2.06 \cdot 10^8 \text{ K} \quad (43)$$

If I am now interested in finding a generic pressure, density and temperature at a specific radius $\tilde{r}$, a possible starting point is to consider:

$$\rho = \rho_c \theta^n \quad (44)$$

with θ given by Eq. (35). The problem now is that ξ is no longer worth π since we are no longer at the outer radius. Here I have to calculate ξ as Eq. (33). I know that: $\xi_1 = \frac{R}{\alpha}$ with R the external radius. So:

$$\tilde{\xi} = \frac{\tilde{r}}{\alpha} = \frac{r}{R} \xi_1 \Big|_{r=3 \cdot 10^8 \text{ m}} \approx 1.178 \quad (45)$$

with $\xi_1 = \pi$ and $R = 8 \cdot 10^8 \text{ m}$. Now I can calculate the $\tilde{\xi}$ evaluated in this newly found ξ , using Eq.(35):

$$\tilde{\theta} = \frac{\sin(\tilde{\xi})}{\tilde{\xi}} \approx 0.78 \text{ rad} \quad (46)$$

And now it is finally possible to calculate the density at a distance of $r = 3 \cdot 10^8 \text{ km}$:

$$\rho(\tilde{r}) = \rho_c \theta^n(\tilde{\xi}) \approx 129.25 \frac{\text{g}}{\text{cm}^3} \quad (47)$$

Clearly I expected it to be smaller than ρ_c . Now, from Eq. (41) it is possible to calculate the pressure at that radius $\tilde{r}$:

$$P(\tilde{r}) = 3.50 \cdot 10^{12} \text{ Pa} \quad (48)$$

The temperature, again from the equation of state, results:

$$T(\tilde{r}) \approx 1255.85 \text{ K} \quad (49)$$

To calculate mass, we have density and radius, so simply:

$$M = \rho V = \rho \frac{4}{3} \pi R^3 = 1.07 \cdot 10^{32} \text{ kg} \approx 54 M_\odot \quad (50)$$

We note that for $n=1$ it makes no sense to write a mass radius relation. Indeed:

$$M = 4\pi g_n \left[\frac{k(n+1)}{4\pi G} \right]^{\frac{n}{n-1}} \left(\frac{R}{\xi_n} \right)^{\frac{3-n}{1-n}} \quad (51)$$

and for $n=1$ the radius does not depend on ρ . Therefore with $n=1$, the above equation is not defined.

Since we know how much ξ is at the surface of radius R , (i.e. π) then it is easy to calculate the quantities at the surface. Indeed, as before:

$$\rho = \rho_c \theta^n \quad (52)$$

On the surface the values are:

$$\theta(\pi) = 0 \quad (53)$$

In general the surface of a polytropic star at $\xi = \xi_1$, where $\theta = 0$, the density and pressure go to zero. So, the density, pressure and temperature, for a polytropic sphere are all zero. The gradient generally is defined as $X_a - X_b/D$ where D is the distance between the two points a and b (in my case $8 \cdot 10^5$ km). Therefore, the gradients are:

$$\begin{aligned} \frac{\rho_s - \rho_c}{R} &\approx 2.1 \cdot 10^{-9} \frac{\text{g}}{\text{cm}^4} \\ \frac{P_s - P_c}{R} &\approx 9.2 \cdot 10^6 \frac{\text{Pa}}{\text{cm}} \\ \frac{T_s - T_c}{R} &\approx 2.6 \cdot 10^{-3} \frac{\text{K}}{\text{cm}} \end{aligned} \quad (54)$$

Exercise 3

A Red Giant star of radius $100 R_\odot$ is in a state where the H in its centre has been consumed, but the He has not yet started to burn out. The energy source is the combustion in the H layer, between $1.8 \cdot 10^9$ cm and $2.0 \cdot 10^9$ cm. In this layer, $\rho = 50 \text{ g/cm}^3$, $T_7 = 5$, $X = 0.5$ and $Z = 0.001$. Calculate the luminosity generated and the T_{eff} of the star.

Solution

Typically, I can consider a red giant as a star where the CNO chain process dominates inside. This can be demonstrated since $q_{pp} \ll q_{CNO}$. Indeed,

$$q_{pp} = 2.4 \cdot 10^6 X^2 \left(\frac{\rho}{1 \text{ gcm}^3} \right) \left(\frac{T}{10^6 \text{ K}} \right)^{-2/3} \exp \left(-\frac{33.8}{(T/10^6 \text{ K})^{1/3}} \right) \frac{\text{erg}}{\text{g s}} \quad (55)$$

and the numerical value is approximately $q_{pp} \approx 230 \frac{\text{erg}}{\text{g s}}$. I now calculate q_{CNO} and expect to find a much larger value.

One of the equation stellar structure bound the luminosity and the energy generation rate:

$$\frac{dL}{dm} = q_{nuc} = q_{CNO} \quad (56)$$

The energy generation rate is defined as:

$$q_{CNO} = 8.7 \cdot 10^{27} X X_{CNO} \left(\frac{\rho}{1 \text{ gcm}^{-3}} \right) \left(\frac{T}{10^6 \text{ K}} \right)^{-2/3} \exp \left(-\frac{152}{(T/10^6 \text{ K})^{1/3}} \right) \text{erg g}^{-1} \text{s}^{-1} \quad (57)$$

where X_{CNO} is the total mass fraction of C, N and O. This is roughly $Z/2$, where Z is the total mass fraction of metals ($Z = 0.001$).

Now, the problem is that $\frac{dL}{dm}$ it is not very convenient to use it, but it is better to rewrite it according to the radius (a piece of data I have available), so:

$$\frac{dL}{dm} = \frac{dL}{dr} \frac{dr}{dm} = \frac{dL}{dr} \frac{1}{4\pi r^2 \rho} = q_{CNO} \quad (58)$$

where in the second passage I used the relationship between mass and density (another equations of stellar structure). By integrating between a radius R_1 and R_2 (given by the problem) I obtain:

$$L = \frac{4\pi}{3} \rho (R_2^3 - R_1^3) q_{CNO} \quad (59)$$

From the data that the problem provides it is possible to obtain q_{CNO} and in particular its value is:

$$q_{CNO} = 9.66 \cdot 10^6 \text{ erg g}^{-1}\text{s}^{-1} \quad (60)$$

As expected $q_{pp} \ll q_{CNO}$. So, the luminosity is:

$$L = 4.388 \cdot 10^{36} \text{ erg s}^{-1} = 4.388 \cdot 10^{29} \text{ J s}^{-1} \quad (61)$$

The effective temperature can be obtained from the generic brightness formula:

$$L = 4\pi R^2 \sigma_{SB} T_{eff}^4 \implies T_{eff} = \left(\frac{L}{4\pi R^2 \sigma_{SB}} \right)^{1/4} = 3364 \text{ K} \quad (62)$$

This result it's supported by the fact that the $T_{eff,\odot} \approx 5400 \text{ K}$, and since here we have a Red Giant, typically it has a lower temperature.

Exercise 4

The core of a star is composed of 90% H and 10% He. Assuming that the material is fully ionised and that the density in the core is $\rho = 150 \text{ g/cm}^3$. Estimate the distance required in order that the radiation field intensity decays by a factor e , if the source of opacity is scattering by free electrons ($\sigma_T \approx 0.6 \cdot 10^{-24} \text{ cm}^2$).

Solution

The efficiency of the medium to absorb a photon can be defined by introducing a typical scale length, which is the mean free path of a photon in the medium l_{mfp} . The mean free path is the average distance that a photon travels in the medium before being absorbed, and not deflected since we are not considering scattering. Let L be the typical size of the medium, there are two cases:

- If $l_{mfp} \gg L$ the medium is said to be optically thin, since usually a photon is not absorbed. In fact, if it is optically thin it's possible to see its internal structure when illuminated.
- If $l_{mfp} \ll L$ the medium is optically thick or opaque, since it does not let the light that hits it pass.

Furthermore, the mean free path l_{mfp} depends on the frequency ν of the photon, so a medium can be optically thin at some frequencies and optically thick at others (think for example of dust or gas for an accretion disk). I can define the optical thickness as the ratio between the length of the medium and the mean free path:

$$\tau_\nu = \frac{L}{l_{mfp}} \quad (63)$$

and a medium is thick if $\tau_\nu \gg 1$ and thin if $\tau_\nu \ll 1$. In radiative transport, usually its inverse, called the **extinction coefficient**, is used instead of the mean free path:

$$\alpha_\nu = \frac{1}{l_{mfp}} \quad (64)$$

The optical thickness between two points (s_0, s_1) inside a medium (or even a point inside and one outside) is defined instead by:

$$\tau_\nu(s_0, s_1) = \int_{s_0}^{s_1} \alpha_\nu(s) ds \quad (65)$$

Given a set of particles of mass density ρ , the extinction coefficient is:

$$\alpha_\nu = \rho \mathcal{K}_\nu \quad (66)$$

with $\mathcal{K}_\nu$ the mass-weighted opacity i.e.:

$$\mathcal{K}_\nu = \frac{\sigma_\nu}{m} \left[\frac{\text{cm}^2}{\text{g}} \right] \quad (67)$$

where σ_ν is the cross section at frequency ν . Combining Eq.(66) with Eq.(67) we obtain that:

$$\alpha_\nu = n \sigma_\nu \quad (68)$$

where n is the volume number density. This way I can rewrite the opacity as follows:

$$\mathcal{K}_\nu \rho = n \sigma_\nu \quad (69)$$

The absorption is therefore dictated by α_ν , while the equivalent for the emission is the **emission coefficient** or **emissivity** and represents the energy emitted per unit of time, frequency, solid angle and volume, and is denoted by j_ν :

$$[j_\nu] = \left[\frac{\text{erg}}{\text{s Hz sr cm}^3} \right] \quad (70)$$

We are finally ready to derive the radiative transport equation. In vacuum (therefore in the absence of absorption and emission) we have seen that:

$$\frac{dI_\nu}{ds} = 0 \quad (71)$$

with s the line parameter of the photon trajectory. This is due to the fact that if a photon travels freely in the universe without encountering any obstacles, it cannot lose energy, so the intensity of the radiation emitted remains constant. If the radiation is absorbed then there will be a loss of radiation intensity, and an increase in the case of emission. For this reason it must be true that:

$$\frac{dI_\nu}{ds} = -\alpha_\nu(s) I_\nu + j_\nu \quad (72)$$

where on the right I_ν is the measured intensity. This is **the equation for radiative transport in the absence of scattering**. For simplicity we will not deal with the emissivity term.

At this point it is really possible to solve the radiative transport equation, thus obtaining:

$$I = I_0 e^{-\tau} \quad (73)$$

Now, if we are interested in estimating the distance at which the radiation intensity has attenuated by a factor e , in formulas

$$\frac{I}{I_0} = e \implies 1 = \tau = \mathcal{K}_\nu \rho s \quad (74)$$

where it was enough to take the logarithm of Eq. (73). The distance is therefore:

$$s = \frac{1}{\mathcal{K}_\nu \rho} = 0.0195 \text{ cm} \quad (75)$$

Where for opacity I used Eq. (69), calculated as:

$$\mathcal{K}_\nu = \frac{\sigma_\nu}{\rho} \cdot n = \frac{\sigma_\nu}{\rho} \frac{\rho}{\mu_e m_p} = \frac{\sigma_\nu}{\mu_e m_p} = 0.34 \frac{\text{cm}^2}{\text{g}} \quad (76)$$

and μ_e is:

$$\mu_e = \left(X + \frac{2Y}{4} + \frac{1}{2}(1 - X - Y) \right)^{-1} = \frac{2}{X + 1} = \frac{20}{19} \quad (77)$$

Exercise 5

The quartic Eddington equation can be written as a function of a mass M that is a combination of natural constants. Find the expression for this mass and calculate it. Express the Chandrasekhar mass as a function of it.

Solution

To solve this exercise it is necessary to remember the so-called *standard model*, which is due to Eddington and is therefore also known as the *Eddington model*. Lets us model the star as a polytrope, a system for which:

$$P = K_P \rho^{\gamma_P} \quad (78)$$

for some constants and K_P and γ_P . It's possible to show that the pressure can be rewritten as:

$$P = \left[\frac{3\mathcal{R}^4(1 - \beta)}{a\mu^4\beta^4} \right]^{\frac{1}{3}} \rho^{\frac{4}{3}} \quad (79)$$

where $\mathcal{R}$ is the *ideal gas constant*, while β represent the fraction of the pressure contributed by the gas and μ is the mass fractions. Now, from the Lane-Emden equation of index n :

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n \quad (80)$$

with the boundary conditions: $\theta = 1$ and $\frac{d\theta}{d\xi} = 0$ at $\xi = 0$, it's possible to find a relation between mass and radius. In particular, the total mass M of a polytropic star is given by:

$$M = \int_0^R 4\pi r^2 \rho dr = 4\pi \alpha \rho_c \int_0^{\xi_1} \xi^2 \theta^n d\xi \quad (81)$$

And from Eq. (80):

$$M = -4\pi \alpha^3 \rho_c \int_0^{\xi_1} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) d\xi = -4\pi \alpha^3 \rho_c \xi_1^2 \left(\frac{d\theta}{d\xi} \right)_{\xi_1} \quad (82)$$

From here, it's possible to derive the relationship between mass and radius:

$$M = 4\pi g_n \left[\frac{K_P(n+1)}{4\pi G} \right]^{\frac{n}{n-1}} \left(\frac{R}{\xi_n} \right)^{\frac{3-n}{1-n}} \quad (83)$$

In the case $n = 3$ the mass does not depend on the radius. This thing is of fundamental importance. All polytropic spheres with $n = 3$ have the same mass. This suggests that all those stars that are well approximated by a polytropic sphere with $n = 3$ then have the same mass! This case applies mainly to high-mass **radioactive stars** ($\approx M_\odot$), but what is this mass worth? We will clarify the limit value on mass in a moment.

The values of the constants $M_n = \xi_1^2 \left(\frac{d\theta}{d\xi} \Big|_{\xi_1} \right)$ vary with the polytropic index n . Knowing that for $n = 3$ is the special case where the mass becomes independent of radius and is uniquely determined by K , it's possible to write:

$$M = 4\pi M_3 \left(\frac{K_P}{\pi G} \right)^{\frac{3}{2}} \quad (84)$$

Substituting K_P from Eq. (79), we obtain the quartic equation in the form:

$$1 - \beta = \left(\frac{M}{M_*} \right)^2 \mu^2 \beta^4 \quad (85)$$

where

$$M_* = \frac{4\sqrt{3} M_3 \mathcal{R}^2}{\sqrt{\pi a G^3}} \quad (86)$$

With $a = 8\pi^5 k^4 / (15c^3 h^3)$ and $R = k/m_H$, I have:

$$M_* = \frac{3\sqrt{10} M_3}{\pi^3} \left(\frac{hc}{Gm_H^{4/3}} \right)^{\frac{3}{2}} = 18.3 M_\odot \quad (87)$$

The existence of an upper limit to the mass of degenerate stars was first found by Chandrasekhar in 1931 and hence the upper limit bears his name, M_{Ch} :

$$M_{Ch} = \frac{M_3 \sqrt{1.5}}{4\pi} \left(\frac{hc}{Gm_H^{4/3}} \right)^{3/2} \mu_e^{-2} \quad (88)$$

Inserting the values of constants:

$$M_{Ch} = 5.83 \mu_e^{-2} M_\odot \quad (89)$$

which yields for $\mu_e = 2$ a limiting mass of $1.46 M_\odot$. In our case, the M_{Ch} can be express as:

$$M_{Ch} = \frac{\pi^2}{8\sqrt{15}} \mu_e^{-2} M_* \quad (90)$$

Exercise 6

Calculate the classical distance of maximum approach for two protons of energy 2 keV. Determine the Gamow Peak energy and estimate the probability of protons penetrating the Coulomb barrier. Repeat the calculations for two He nuclei with the same energy of approximation.

Solution

Let r_0 be the distance of closest approach. As the two protons approach each other, the potential energy rises. The distance of closest approach is the distance at which all the initial kinetic energy is transformed to potential energy, so:

$$K_{before} = \frac{1}{2}m_p v^2 + \frac{1}{2}m_p v^2 = \frac{Z_i Z_j e^2}{4\pi\epsilon_0 r_0} = U_{final} \quad (91)$$

Solve for r_0 :

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{e^2}{K_{before}} = \frac{1}{4\pi} \frac{(1.6 \cdot 10^{-19})^2}{8.85 \cdot 10^{-12} \cdot 3.2 \cdot 10^{-16}} = 7.19 \cdot 10^{-13} \approx 700 \text{ fm.} \quad (92)$$

where here $Z_i Z_j = 1$ because I have only proton. The Gamow peak is determined by the function:

$$f(E) = e^{-\frac{E}{k_B T}} e^{-\frac{b}{\sqrt{E}}} \quad (93)$$

where b depends only on the charges and masses of the nuclei involved in the reaction. Taking the derivative of Eq.(93) and setting it equal to zero we obtain the energy at the Gamow peak:

$$-\frac{1}{k_B T} + \frac{b}{2E_{peak}^{3/2}} = 0 \implies E_{peak} = \left(\frac{b k_B T}{2} \right)^{2/3} \quad (94)$$

The parameter b is defined as:

$$b = 2^{3/2} \pi^2 \frac{\mu_{red}^{1/2} m_H^{1/2} Z_i Z_j e^2}{h} = 0.0013 \mu_{red}^{1/2} Z_i Z_j \text{ (erg)}^{1/2} \quad (95)$$

where $\mu_{red} = m_i m_j / (m_i + m_j)$. Clearly the temperature is related to the kinetic energy via $3/2 k_B T$. So in my case:

$$T \approx 1.57 \cdot 10^7 K \quad (96)$$

Putting b inside the peak energy I get:

$$E_0 = 1.22 \left[Z_i^2 Z_j^2 \mu_{red} \left(\frac{T}{10^6 [K]} \right)^2 \right]^{1/3} \text{ keV} \quad (97)$$

In my case I'm dealing with protons, so $Z_1 = Z_2 = 1$ and $\mu_{red} = 1/2$. I therefore obtain that:

$$E_0 = 6.01 \text{ keV} \quad (98)$$

Finally, if the probability of penetrating the Coulomb barrier for two generical nuclei with charge numbers Z_i and Z_j is:

$$P(E) = \exp \left(-\frac{b}{\sqrt{E}} \right) \quad (99)$$

with

$$b = 31.28 Z_i Z_j \mu_{red}^{1/2} (keV)^{-\frac{1}{2}} \quad (100)$$

and $E = 2keV$, I get a probability of:

$$P(E) = 1.6 \cdot 10^{-7} \quad (101)$$

That is, for every 10 million protons, one manages to overcome the barrier. Now, I should do the same thing with two helium nuclei. The only differences compared to before are the atomic number $Z_1 = Z_2 = 2$, and consequently the mass, that is $\mu_{red} = 2$. Furthermore also r_0 change because now I have $Z_i Z_j = 4$ so $r_0 = 2800$ fm. The energy at the Gamow peak, from Eq.(97) it results to be:

$$E_0 = 24.28 \text{ keV} \quad (102)$$

and a probability of:

$$P(E) = 4.58 \cdot 10^{-55} \quad (103)$$

Exercise 7

If the proton-proton chain occurs at a temperature corresponding to $k_B T = 1.4$ keV, estimate the Gamow Peak width of the fusion window. Determine the temperature dependence of the reaction rate. Discuss how this dependence changes by increasing the temperature.

Solution

The width of the Gamow peak is:

$$\Delta = \frac{4}{\sqrt{3}} (E_0 k_B T)^{1/2} \quad (104)$$

Using Eq.(97) given from the previous exercise, we obtain:

$$E_0 = 6.23 \text{ keV} \quad (105)$$

and so, the width is:

$$\Delta = 6.82 \text{ keV} \quad (106)$$

Having an approximate understanding of the relationship between reaction rate and temperature in the vicinity of the ignition temperature proves advantageous. This understanding is attained by an approximate assessment of the integral within the formula governing the rate coefficient. As an initial step in this approximation, any potential fluctuations in the $S(E)$ factor across the Gamow peak are disregarded, with the factor instead being equated to a constant value, denoted as $S(E_0)$. Consequently, the reaction rate coefficient is approximated as follows:

$$R_{ijk} = \frac{1}{(\pi \mu_{red} m_H)^{1/2}} \left(\frac{2}{k_B T} \right)^{3/2} S(E_0) \int_0^{\infty} \exp \left(-\frac{E}{k_B T} - \frac{b}{\sqrt{E}} \right) dE \quad (107)$$

The maximum value of the integrand occurs when $E = E_0$, and is given by

$$I_{max} = \exp \left(-\frac{3E_0}{k_B T} \right) \equiv e^{-\tau} \quad (108)$$

where I define:

$$\tau = \frac{3E_0}{k_B T} = 42.46 \left[Z_i^2 Z_j^2 \mu_{red} \left(\frac{T}{10^6 [K]} \right)^{-1} \right]^{\frac{1}{3}} \quad (109)$$

The second step in the approximation is to approximate the exponential factor in the integral by a Gaussian of width Δ :

$$\exp \left(-\frac{E}{k_B T} - \frac{b}{\sqrt{E}} \right) \approx I_{max} \exp \left[-\left(\frac{E - E_0}{\Delta/2} \right)^2 \right] \quad (110)$$

The width Δ is generally chosen by picking the value such that the second derivatives of the exact and approximate forms for the integrand are equal at $E = E_0$. From here it is possible to arrive at Eq.(104) given at the beginning.

The final step in the approximation is to change the limits of integration from 0 to ∞ to $-\infty$ to ∞ . This is not a bad approximation because the vast majority of the power in the Gaussian occurs at positive energies, and if the limits are $-\infty$ to ∞ , the integral can be done exactly:

$$\int_{-\infty}^{\infty} \exp \left[-\left(\frac{E - E_0}{\Delta/2} \right)^2 \right] dE = \frac{\sqrt{\pi}}{2} \Delta \quad (111)$$

With this approximation complete, we can write the reaction rate coefficient as:

$$R_{ijk} = I_{max} \left(\frac{2}{\mu_{red} m_H} \right)^{1/2} \frac{\Delta}{(k_B T)^{3/2}} S(E_0) \quad (112)$$

We can rewrite this in terms of τ by substituting in for Δ and $k_B T$ in terms of τ . Doing so and simplifying a great deal produces:

$$R_{ijk} = \frac{4}{3^{5/2} \pi^2} \frac{h}{\mu_{red} m_H Z_i Z_j e^2} S(E_0) \tau^2 e^{-\tau} \quad (113)$$

All the temperature-dependence is encapsulated in the $\tau^2 e^{-\tau}$ term. The factor τ itself varies as:

$$\tau \propto \frac{E_0}{T} \propto T^{-1/3} \quad (114)$$

Approximating the reaction rate using a power law in temperature, where R_{ijk} varies with T raised to a power ν , proves to be beneficial. While the relationship isn't strictly a power law due to the presence of an exponential term in τ , it can be treated as such near a specific temperature, T_0 , where τ exhibits a particular dependency on T through $\tau = \tau_0 (T/T_0)^{(-1/3)}$. Understanding this approximation involves recognizing that a power law manifests as a straight line when plotted on a logarithmic scale. Effectively, fitting to a power law is analogous to determining the slope at a specific point on the logarithmic plot. Thus:

$$\nu = \frac{d \ln(R_{ijk})}{d \ln(T)} \quad (115)$$

Since $R_{ijk} \propto \tau^2 e^{-\tau}$,

$$\ln(R_{ijk}) = 2 \ln(\tau) - \tau + const = -\frac{2}{3} \ln(T) - \tau_0 \left(\frac{T}{T_0} \right)^{-\frac{1}{3}} + const \quad (116)$$

Taking the derivative:

$$\nu = \frac{\ln(R_{ijk})}{d \ln(T)} = -\frac{2}{3} - \tau_0 T_0^{1/3} \frac{d}{d \ln(T)} T^{-1/3} = -\frac{2}{3} - \tau_0 T_0^{1/3} T \frac{d}{dT} T^{-1/3} = \frac{\tau}{3} - \frac{2}{3} \quad (117)$$

This lets us approximate the behavior of R_{ijk} as a powerlaw:

$$R_{ijk} = R_{0,ijk} T^{(\tau-2)/3} \quad (118)$$

However, plotting this function and remembering that tau depends on temperature, it is possible to observe an increase up to a maximum, and then decrease indefinitely until zero. In other words, for $T \rightarrow \infty$ then $R_{ijk} \rightarrow 0$.

Appendix A

Virial Theorem

To derive the virial theorem, we will start by taking both sides of the Lagrangian equation of hydrostatic balance (32) and multiplying by the volume $V = 4\pi r^3/3$ interior to some radius r :

$$VdP = -\frac{1}{3} \frac{Gm dm}{r}$$

Next we integrate both sides from the center of the star to some radius r where the mass enclosed is $m(r)$ and the pressure is $P(r)$:

$$\int_{P(0)}^{P(r)} VdP = -\frac{1}{3} \int_0^{m(r)} \frac{Gm' dm'}{r'}$$

Since Gm'/r' is the gravitational potential due to the material of mass m' inside radius r' , the integrand $(Gm'/r')dm'$ just represents the gravitational potential energy of the shell of material of mass dm' . Thus the integrand on the right-hand side is just the gravitational potential energy of each mass shell. When this is integrated over all the mass interior to some radius, the result is the total gravitational potential energy of the gas inside this radius. Thus we define:

$$\Omega(r) = - \int_0^{m(r)} \frac{Gm' dm'}{r'}$$

to the gravitational energy of the gas inside radius r . Turning back to the left-hand side, I can integrate by parts:

$$\int_{P(0)}^{P(r)} VdP = [PV]_0^r - \int_0^{V(r)} PdV = [PV]_r - \int_0^{V(r)} PdV$$

In the second step, I dropped PV evaluated at $r' = 0$, because $V(0) = 0$. Since $dm = 4\pi r^2 \rho dr$ I can write $dV = dm/\rho$ and this changes the integral to:

$$\int_0^{V(r)} PdV = \int_0^{m(r)} \frac{P}{\rho} dm$$

Putting everything together, we arrive at our form of the virial theorem:

$$[PV]_r - \int_0^{m(r)} \frac{P}{\rho} dm = \frac{\Omega(r)}{3}$$

if we opt to apply this theorem at the outer radius of the star, setting $r = R$, the first term vanishes due to the negligible surface pressure, thus yielding:

$$\int_0^M \frac{P}{\rho} dm = -\frac{1}{3}\Omega$$

where Ω is the total gravitational energy of the star. Now, consider a ideal gas, I know that:

$$P = \frac{\rho k_B T}{\mu m_H} = \frac{\mathcal{R}}{\mu} \rho T$$

where μ is the mean mass per particle in the gas, measured in units of the hydrogen mass, and $\mathcal{R} = k_B/m_H$ is the ideal gas constant. Substitute this into the virial theorem, I get:

$$\int_0^M \frac{\mathcal{R} T}{\mu} dm = -\frac{1}{3}\Omega \quad (\text{A.1})$$

For a monatomic ideal gas, the internal energy per particle is $(3/2)k_B T$, so the internal energy per unit mass is $u = (3/2)\mathcal{R} T/\mu$. At the end I obtain:

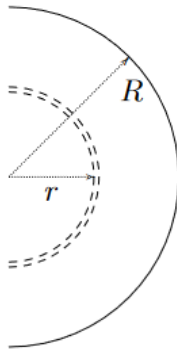
$$\int_0^M \frac{2}{3} u dm = -\frac{1}{3}\Omega \implies U = -\frac{1}{2}\Omega$$

where U is just the total internal energy of the star, i.e. the internal energy per unit mass u summed over all the mass in the star. It tells us that the total internal energy of the star is simply $-(1/2)$ of its gravitational binding energy.

Appendix B

Third stellar structure equations

Energy Production Equation



Consider a spherical shell of thickness dr in a spherically symmetric star of radius R in which energy transport is radial and time variations are unimportant. Let's $L(r)$ the rate of energy flow across sphere of radius r and $L(r + dr)$ the rate of energy flow across sphere of radius $r + dr$. Cause of the thinly shell (see figure on the left), I can write that:

$$dV(r) = 4\pi r^2 dr \quad \text{and} \quad dM(r) = 4\pi r^2 \rho dr$$

Define q_{nuc} as the energy release per unit mass in unit time and hence rate of energy release in the shell is $4\pi r^2 \rho(r) q_{nuc} dr$. As energy is conserved:

$$L(r + dr) = L(r) + 4\pi r^2 \rho(r) q_{nuc} dr$$

and therefore:

$$\frac{L(r + dr) - L(r)}{dr} = 4\pi r^2 \rho(r) q_{nuc}$$

In the limit where $dr \rightarrow 0$ I obtain:

$$\frac{dL}{dr} = 4\pi r^2 \rho(r) q_{nuc} \implies \frac{dL}{dm} = q_{nuc} \quad (\text{B.1})$$